CS-330

Reagan Morgan

Professor Brian Battersby

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Final Project Reflection & Design Decisions

For my 3D scene, I wanted to choose something that would challenge me but also be obtainable within the parameters and scope laid out by the rubric. This led me to choose a scene with a lot of basic primitive shapes that I could remodel in OpenGL. For example, one of the best implementations of using primitive shapes to recreate a complex object within the scene was my pipe implementation. At first, I was having trouble trying to figure out how I could make my pipe bend without looking too sharp at the edges. Eventually, after a bit of fine-tuning, I came to the realization that I could use spheres on the edge of each long, thin cylinder that intersected at around 1/3 of the sphere on two sides in order to give the pipes a realistic looking bend. While I could have added even more detail onto these pipes (like small fasteners made from tori or extremely small cylinders), I realized that this amount of detail would not lead me any closer to the true destination of this project, to make a deliverable that has enough detail to be recognizable as the final image, while still being relatively simplistic in terms of detail. For me, balancing this was one of the hardest parts of this project. I also had issues with certain parts of the image not being easy to recreate given the predefined primatives. An example of this would be the shingles on the roof. They intersect at each corner of the building along their smallest edge length. I was hoping I would be able to reproduce this at first using boxes rotated at an angle around the x-axis but then realized I would not be able to get those connections within the dimensions of my building without edges poking out of each box shape. I struggled with this at

first. I concluded that the closest I could get was using prisms that were half in and half out of the top roof structure. This still wasn't perfect as the edges were not connected between each prism, but it negated the issue of overlapping that the boxes had. Ultimately, if I were to have a bit more time and less constraints with the knowledge I have now, I probably would have tried to create a trapezoidal prism so that way I could use a similar technique to the prisms I implemented, but with the edges being able to connect. The other main issue I ran into was implementing the lighting the way I wanted it to be with the lantern. I first had issues with the lighting techniques I tried on the paper lantern not showing up the way I wanted them to with the pointlight in the center of the transparent object. This led to immense frustration when trying to figure out how to implement this. Eventually, I had to use three point lights coming from the left, right and front of my lantern, along with making the lantern opaque, to get an effect like what I wanted. This led to another issue however of the light being super bright on sides of the roof where it should have been dark except for the directional light I added to simulate moonlight. I was never able to fix this issue, but if I had more time, this would be one of the first things I try to fix.

The navigation within my scene is fairly simple. It uses the ASWD keys to move left, right, up, and down, respectively. It also uses the scroll wheel to control the speed at which the camera moves across the screen, and the cursor is bound to the center to simulate looking around the scene. I also implemented an orthographic front camera view to the O key, as well as a perspective view from the top to the P key. This allowed me to see the scene at different angles while I was working to make sure each object looked normal from all sides.

The biggest thing I would fix in my code is creating functions. I admittedly fell short of this requirement within my code that causes my code to be a bit long winded and repetitive. I tried to reduce this mistake by implementing as many separations as I could within the code, like

you see in the PrepareScene() function where objects are separated out by whether they are planes, objects relating to the building (walls, roofs, flooring), exterior components like the lantern and gas tank, or interior components like the chairs and counter. Things like this are why it is so important to read through all requirements that are asked of you before the last moment. While I missed the mark on this part of the assignment, I will take this failure into my future classes and future life as a developer.